EXP 7: LOGISTIC REGRESSION

NAME: SAMUEL I

REG NO:192312584

AIM:

To implement the Logistic Regression algorithm in Python and classify the given dataset with performance evaluation.

PROCEDURE

1. Import the required libraries and load the dataset into the program.
2. Split the dataset into training data and testing data.
3. Train the Logistic Regression model using the training dataset.
4. Predict the class labels for the testing dataset.
5. Display the confusion matrix and accuracy to evaluate the performance of the model.

PROGRAM:

```
# Logistic Regression using Python


from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LogisticRegression

from sklearn.metrics import confusion_matrix, accuracy_score


# Load dataset

iris = load_iris()

X = iris.data

y = iris.target


# Split dataset

X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y, test_size=0.3, random_state=42
)

# Create and train Logistic Regression model
model = LogisticRegression(max_iter=200)
model.fit(X_train, y_train)

# Predict test data
y_pred = model.predict(X_test)

# Display results
print("Predicted Values:")
print(y_pred)

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

print("\nAccuracy:")
print(accuracy_score(y_test, y_pred) * 100, "%")
```

OUTPUT:

```
Predicted Values:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 1
 0 0 0 2 1 1 0 0]

Confusion Matrix:
[[19  0  0]
 [ 0 13  0]
 [ 0  0 13]]

Accuracy:
100.0 %
```

RESULT:

The Logistic Regression algorithm was successfully implemented in Python. The model was trained and tested on the dataset, and its performance was evaluated using the confusion matrix and accuracy, achieving effective classification results.